25900516 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC)- MG3VACMAT200 - MASTERING PROBLEM SOLVING THROUGH VEDIC MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 15 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Each question must be solved using the relevant Vedic Sutras with proper steps

Answer any **5** questions

Each question carries **2** marks

1. Using Ekadhikena Purvena method, find the sum of 95,37,87,99 [S] / [CO2]
2. Write the importance of zero in the complement of divisor. [K] / [CO1]
3. Multiply 2487 by 9999. [S] / [CO2]
4. Why is the Beejank Method useful in cube root calculations up to 7 digits. [U] / [CO1]
5. If A : B = 1 : 2 , B : C = 3 : 4 , C : D = 6 : 9 and D : E = 12 : 16, find A : B : C : D : E [An] / [CO5]
6. A man buys a toy for ₹25 and sells it for ₹30 . Find his gain percent. [A] / [CO6]
7. Solve $\frac{2}{x+1} + \frac{3}{x+2} = 0$, and find the value of x. [A] / [CO7]
8. Solve: $3x + 2y = 5$
   $6x + 8y = 10$ [U] / [CO8]

[2x5 = 10]

Part B

Short Essay Questions

Each question must be solved using the relevant Vedic Sutras with proper steps

Answer any **4** questions

Each question carries **5** marks

9. Multiply 9604 by 111. [S] / [CO2]

10. Using Nikhilam Navatascaramam method, subtract 45993 from 87453 [U] / [CO1]

11. Apply Duplex Method to find the square of 45612 [A] / [CO5]

12. At what rate per cent compound interest does a sum of money become nine-fold in 2 years? [A] / [CO5]

13. Find the product of $2x^2 + 3y$ and $2x^2 + 5y$ using Urdhva Tiryagbhyam. [A] / [CO7]

14. Solve $\frac{3}{x+1} + \frac{4}{x+2} = \frac{7}{x+3}$ using Paravartya method. [An] / [CO7]

[5x4 = 20]

Part C

Essay Questions

Each question must be solved using the relevant Vedic Sutras with proper steps

Answer any **2** questions

Each question carries **10** marks

15. Subtract 122245 from 513323 using digit separator method. [S] / [CO2]

16. Determine whether 62932451 is divisible by 139 using the Vedic method of osculators [U] / [CO1]

17. The population of a town is 10,000. It increases by 10% during the first year. During the second year, it decreases by 20% and increased by 30% during the third year. What is the population after 3 years? [A] / [CO3]

18. Multiply the following using Urdhva Tiryagbhyam: [U] / [CO1]
    (a) $(2x^2 - 3x + 4) \times (x^2 + 5x - 6)$.
    (b) $(4x^2 - 1) \times (x^2 - 3x + 7)$.

[10x2 = 20]